Neutral Hubs Beyond Direct Transmission: How Homophily and the Spiral of Silence Shape Downstream Posting Dynamics

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Abstract. As web communication platforms expand, hub users play critical roles in online opinion shaping, yet the effect of neutrally positioned hubs on polarized posting dynamics remains poorly understood. Using an agent-based model on heavy-tailed adaptive social networks, we analyze how neutral hubs shape opinion sharing. Active neutral hubs consistently promote neutral-camp posting while suppressing moderately opinionated posting. However, their effect on the ideologically distant extreme camp is indirect and can reverse across underlying model conditions. We trace this sign reversal to a two-layer mechanism: network-level homophily, which regulates extreme agents' exposure to moderately opinionated content, and a spiral-of-silence expression threshold, which governs their response. Our findings demonstrate that neutral hubs operate beyond direct transmission; by regulating content flows, they induce qualitatively different downstream effects through the interplay of structural filtering and individual-level evaluation.

Keywords: Social networks, Opinion dynamics, Agent-based simulation, Social media, Computational Social Science

1 Introduction

Social media platforms have become primary venues for public discourse, deeply shaping opinion formation [23]. Unlike traditional face-to-face communication, people communicate and process information through individually structured information environments on these platforms [2]. The collective dynamics emerging from these localized patterns of communication have therefore become an important subject of study. As social media platforms have become a fundamental infrastructure for information processing and public discourse, and digital platforms are increasingly being considered as means of expanding collective deliberation and democratic participation, understanding the dynamics they generate

is increasingly essential to understanding the conditions under which contemporary societies communicate, deliberate, and form opinions.

Among these distinctive features of social media, hub users, such as influencers, can reach a disproportionately large portion of a platform's population, giving them substantial structural influence. While the role of these hubs in information diffusion is well studied [21, 1, 4], their consequences for polarized opinion and posting dynamics remain less understood, particularly when highly connected users themselves occupy neutral positions rather than extreme ones. In particular, as a topic becomes increasingly controversial among society, popular figures with neutral stances along the relevant opinion dimension may enter the discussion, yet the consequences of their engagement remain largely unexplored.

Therefore, we investigate how the activity of neutral hubs shapes polarized posting dynamics. Using an agent-based model integrating bounded-confidence [10, 16] and spiral-of-silence-inspired [22] mechanisms on heavy-tailed adaptive social networks, we find that neutral hubs consistently boost posting in the aligned neutral camp while suppressing it in the neighboring moderately opinionated camp. In contrast, their effect on the distant extreme camp is indirect and can reverse depending on underlying model conditions. Rather than reaching extreme agents directly, neutral hubs alter the vocality and information flow of moderately opinionated agents. This downstream process is filtered at two levels: network-level homophily determines how much aligned content an agent is exposed to, while an agent-level expression threshold determines whether that content is comfortable enough to voice. Their interaction ultimately determines whether neutral-hub activity facilitates or suppresses extreme posting.

By examining vocality alongside opinion dynamics, we show that neutral hubs alter the visibility of neighboring camps and shift downstream dynamics, effects invisible when tracking internal opinions alone. Our results suggest that the role of neutral hubs as stabilizing bridges or brokers, often expected to improve polarized information environments, does not necessarily yield linear or intuitively integrative outcomes. Our results challenge the intuitive view of neutral hubs as stabilizing bridges or brokers in polarized environments. Instead, their activity propagates indirectly by reshaping who becomes vocal within the network. These findings underscore that understanding social media dynamics requires attention not only to how opinions and ties evolve, but also to how patterns of posting and silence alter information flows across the platform.

2 Related Work

While early social network models focused on static topologies [26], recent work models systems where topology coevolves with agent opinions and behavior. Building on the classic voter model [8], Vazquez et al. [25] used an agent-based model (ABM) to show that coupled opinion updating and edge rewiring induce a phase transition between connected and fragmented networks. Applying this coevolutionary framework, Kozma and Barrat [19] integrated bounded confidence [10, 16], where agents interact only within a tolerance threshold, demon-

strating that homophilic rewiring drives network fragmentation. Extending this to modern web platforms, Sasahara et al. [24] confirmed that social influence combined with unfollowing reproduces the political polarization observed in empirical social media data [9].

Whether such bridging roles are inherently stabilizing, however, is far from settled on social media. Classic network theory emphasizes that bridging ties—whether as weak ties [15] or structural brokers [6] with high betweenness centrality [12]—foster connectivity across communities. On social media, however, this integrative function is conditional: follower count does not directly equate to influence [7], and hub impacts depend heavily on opinion dynamics. Using ABMs, Helfmann et al. [17] showed that extreme hubs gain more followers, while Betts et al. [4] found that a neutral influencer's effect inverts with trust: narrow trust backfires and increases polarization, whereas broad trust reduces it. Thus, neutral hubs do not automatically act as stabilizing mediators.

Most existing studies focus on opinion positions themselves—either individual attitudes [17] or aggregate polarization [4, 24]—as their primary outcome. On social media, however, private opinions remain unobservable; only expressed content is visible, making vocality essential to understanding public discourse. Grounded in spiral-of-silence theory [22], where individuals modulate expression based on perceived opinion climate [20], we treat posting behavior, or vocality, as our primary outcome, examining how neutral hubs' activity reshapes observable opinion expression across groups.

3 Model

3.1 Agent-Based Model

We simulate opinion and posting dynamics on a directed graph $G = (V, E)$, where $V = \{1, 2, \ldots, N\}$ is a set of N agents representing social media users, and $E \subseteq V \times V$ is the set of directed follow edges. An edge $(i, j) \in E$ means agent i follows agent j, represented by the adjacency matrix $\mathbf{A}_{ij} = 1$ if $(i, j) \in E$ and 0 otherwise. This network evolves through the social media behaviors described in Section 3.3 over discrete time steps $t \in \{0, 1, 2, \ldots\}$.

Each agent $i \in V$ holds an opinion $O_i(t) \in [-1, 1]$ at time t, with -1 and 1 denoting opposing extreme positions and 0 a neutral stance. We divide $[-1, 1]$ into five equal-width bins to categorize agents into three camps based on opinion magnitude: the *neutral* camp ($|O_i(t)| < 0.2$), the *moderately opinionated* camp ($0.2 \leq |O_i(t)| < 0.6$), and the *extreme* camp ($|O_i(t)| \geq 0.6$), with sign indicating ideological alignment. Hereafter, we refer to the moderately opinionated camp as the *moderate* camp. An agent's initial opinion $O_i(0)$, termed its *intrinsic opinion*, represents a latent attitude drawn independently from a Gaussian distribution truncated to $[-1, 1]$.

Beyond its opinion, each agent is characterized by two further internal states, summarized in Table 1: a posting probability $\rho_i(t)$, governing how actively it posts, and a *bounded-confidence threshold* $\epsilon_i(t)$, governing which content from other agents it tolerates well enough to remain connected with.

Table 1: Agent state variables and their value ranges.

State variable	Symbol	Range	Initial value
Opinion	$O_i(t)$	$[-1.0, 1.0]$	$\mathcal{N}(0.0, 0.6^2)$
Posting probability	$\rho_i(t)$	$[0.01, 1.0]$	0.26
Bounded-confidence threshold	$\epsilon_i(t)$	$[0.2, 1.0]$	0.8

3.2 Modeling Social Adaptation

Each agent updates its internal state only when it accesses the platform, an event that occurs independently at each time step with access probability P_a. Therefore, every rule introduced below is understood to fire once per access rather than once per step; accordingly, t and $t{+}1$ in Eqs. (1)–(4) index agent i's successive accesses, not the global simulation step.

Social influence. Upon accessing the platform, agent i reads all posts in its feed $F_i(t)$ and updates its opinion by averaging those within its tolerance threshold $\epsilon_i(t)$ [13]. As shown in Eq. (1), this average is blended with its intrinsic opinion $O_i(0)$ according to a stubbornness parameter $\mu \in [0, 1]$. If no post falls within $\epsilon_i(t)$, then $O_i(t)$ remains unchanged.

$$O_i(t{+}1) = \mu \cdot O_i(0) + (1 - \mu) \cdot \frac{\sum_{p \in \widehat{F}_i(t)} O(p)}{\left|\widehat{F}_i(t)\right|}. \tag{1}$$

where $\widehat{F}_i(t) = \{p \in F_i(t) : |O(p) - O_i(t)| < \epsilon_i(t)\}$.

Confidence decline and recovery. Bounded confidence $\epsilon_i(t)$ is adjusted at every access toward a target level determined by how much of the feed the agent tolerates, with out-of-tolerance posts narrowing this threshold. Writing $r_i(t)$ for the fraction of i's feed that falls within its current threshold, the target level is defined as

$$\epsilon_i^*(t) = \epsilon_{\text{floor}} + (\epsilon_{\text{ceil}} - \epsilon_{\text{floor}}) \cdot r_i(t), \tag{2}$$

and the threshold is relaxed toward it according to

$$\epsilon_i(t{+}1) = \epsilon_i(t) + \begin{cases} \delta_\epsilon \cdot \big(\epsilon_i^*(t) - \epsilon_i(t)\big) & \text{if } \epsilon_i^*(t) < \epsilon_i(t), \\ \gamma_\epsilon \cdot \big(\epsilon_i^*(t) - \epsilon_i(t)\big) & \text{otherwise,} \end{cases} \tag{3}$$

clipped to $[\epsilon_{\text{floor}}, \epsilon_{\text{ceil}}]$, where δ_ϵ and γ_ϵ ($\delta_\epsilon \gg \gamma_\epsilon$) reflects the well-documented asymmetry by which negative experiences carry greater psychological weight [3].

Social reinforcement and vocality. Whether an agent tolerates a post is conceptually distinct from whether that post emboldens it to express its opinion. A post p_j by agent j is defined as *comfortable* for agent i if $|O_i(t) - O(p_j)| < \kappa$, where the vocal comfort radius κ is a population-wide threshold, distinct from

$\epsilon_i(t)$. While $\epsilon_i(t)$ governs individual tolerance for disagreement, κ, inspired by the spiral of silence [22], determines whether encountered content creates an environment conducive to expression. The latter thus reflects a shared norm of expressive comfort that may vary across topics or groups, through which agents subjectively evaluate the content they encounter in their feeds. Letting $c_i(t)$ denote the fraction of comfortable posts in i's feed, the posting probability relaxes toward a target value determined linearly by $c_i(t)$:

$$\begin{aligned} \rho_i(t{+}1) &= \rho_i(t) + \eta \cdot \big(\rho_i^*(t) - \rho_i(t)\big) \text{ and} \\ \rho_i^*(t) &= \rho_{\min} + (\rho_{\max} - \rho_{\min}) \cdot c_i(t), \end{aligned} \tag{4}$$

where $\eta > 0$ is the posting-probability relaxation rate and the updated value is clipped to $[\rho_{\min}, \rho_{\max}]$. As with confidence decline, this graded formulation ensures that every comfort level yields a distinct equilibrium vocality.

3.3 Modeling Behaviors on Social Media

At each access, agent i creates a post reflecting its current opinion $O_i(t)$ with probability $\rho_i(t)$, which is distributed to all its current followers. Agent i's feed $F_i(t)$ consists of up to S_{fd} recent original or reposted posts from agents it follows. Upon reading its feed, i reposts each post within its bounded-confidence threshold $\epsilon_i(t)$ with probability P_r. Posts outside $\epsilon_i(t)$ are reposted at a much lower rate $P_r^{\text{out}} \ll P_r$, capturing occasional out-of-tolerance amplification, such as quote-posting disagreeable content to react to it.

After reading its feed, agent i collects candidate authors of all posts within its bounded-confidence threshold whom it neither follows nor blocks. With probability P_f, it follows one candidate chosen uniformly at random. An agent can follow at most S_{fol} accounts, reflecting cognitive limits and information overload that constrain effective social influence [14, 27]. Upon reaching this capacity, adding a new follow immediately evicts the currently followed agent whose opinion is farthest from i's own.

When accessing the platform, agent i decides with probability P_u whether to act on out-of-tolerance content. If triggered, i scans its feed for the first post from a followed author whose opinion lies outside $\epsilon_i(t)$; if found, i unfollows and blocks that author. Otherwise, if the feed contains an out-of-tolerance post from an unfollowed author, i blocks one such author at random. Thus, at most one tie is severed or blocked per access. The parameter P_u sets the population's overall propensity to prune disagreeable ties, acting as the model's central knob governing network fragmentation.

Posting probability $\rho_i(t)$ is kept strictly independent of opinion extremity $|O_i(t)|$ throughout the rules above. This ensures that posting and reposting behaviors act as our primary outcomes: social influence is driven by surfaced, vocalized activity rather than unexpressed internal opinions.

Table 2: Model parameters and their default values.

Parameter	Symbol	Default value
Probability of accessing the platform	P_a	0.1
Probability of following	P_f	0.01
Probability of reposting (in/out of tolerance)	P_r, P_r^{out}	0.1, 0.01
Probability of unfollowing/blocking	P_u	0.05 (variable)
Maximum number of followees	S_{fol}	43.9
Feed capacity	S_{fd}	15
Confidence decline/recovery rate	δ_ϵ, γ_ϵ	0.5, 10^{-4}
Bounded confidence floor/ceiling	$\epsilon_{\text{floor}}, \epsilon_{\text{ceil}}$	0.2, 1.0
Vocal comfort radius	κ	0.1 (variable)
Posting-probability relaxation rate	η	0.1
Posting-probability floor/ceiling	$\rho_{\min}, \rho_{\max}$	0.01, 1.0
Stubbornness	μ	0.82

3.4 Modeling the Neutral Hub Intervention

This study exogenously fixes the opinions of a designated set of highly reachable agents. At $t = 0$, we rank all agents by initial follower count—the sole reach mechanism in this model—and select the top $N_{\text{hub}} = 60$ agents as the hub pool. As shown in Table 3, this pool size is conservative relative to the network's high-reach plateau. The pool is partitioned symmetrically into five target positions representing the camps in Section 3.1: ten agents each at $O = \pm 1.0$ (*extreme*), ten each at $O = \pm 0.4$ (*moderate*), and twenty at $O = 0.0$ (*neutral*), the central focus of our study. At $t = 0$, each hub's intrinsic and current opinions are set to its assigned target; we confirmed these opinions change negligibly during runs. Their stubbornness, posting probability, and bounded-confidence dynamics remain unmodified.

The model also supports silencing specific hub categories. A silenced hub cannot post or repost, but continues reading its feed and updating ties as before, leaving its follow dynamics unchanged. Comparing an active-hub condition against an otherwise identical hub-silenced condition isolates the effect attributable directly to the hub's vocal activity. Section 4 details the resulting experimental setups.

4 Experimental Design

The initial follow network is generated via a directed Holme-Kim process [18] ($m = 15$, initial attractiveness $A_0 = 4$ [11], $p_t = 0.05$), analytically yielding a degree exponent $\gamma = 2 + A_0/m \approx 2.267$ to mimic real-world social networks. Table 3 reports its structural properties at $t = 0$ and $t = 40{,}000$, along with the structural position of the $N_{\text{hub}} = 60$ hub pool at each point.

Most parameters in Table 2 and the network generator were fixed via a parameter sweep targeting empirical Twitter/X features: heavy-tailed degree

Table 3: Network structure at $t = 0$ and $t = 40{,}000$ (control, $P_u = 0.05$, $\kappa = 0.1$; 8-seed mean).

	$t = 0$	$t = 40{,}000$
Mean degree $\langle k \rangle$	15.0	9.7
Clustering coefficient	0.14	0.30
Hub in-degree percentile (mean/min)	96.9 / 94.0	94.5 / 45.6
Hub betweenness percentile (mean/min)	70.6 / 29.5	69.4 / 45.1
Hub betweenness mass share	9.8%	16.0%

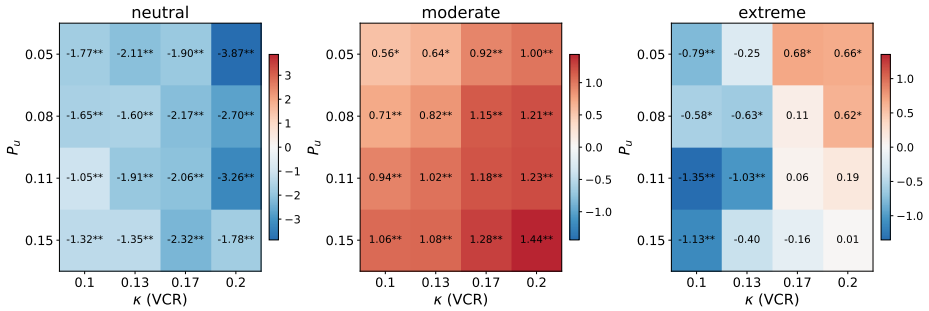


Fig. 1: Per-camp effect of neutral-hub silencing on posting rate (N-silent – control; paired Cohen's d_z; one-sided paired t-test: $* \; p < 0.05$, $** \; p < 0.01$)

distributions [21], high reposting modularity [9], and opinion-weighted posting dynamics [5]. This calibration aligns the model with empirical polarized posting dynamics. In contrast, P_u and κ were excluded from calibration and serve as directly manipulated experimental factors.

Within this calibrated network, we evaluate two conditions (Section 3.4): a *control* condition in which all five hub categories remain active, and an *N-silent* condition in which only the neutral hubs are silenced. Because silencing preserves follow dynamics, any outcome difference between these conditions can be attributed directly to the neutral hubs' vocal activity.

We cross this condition with a full factorial sweep of two parameters: the vocal comfort radius $\kappa \in \{0.10, 0.13, 0.17, 0.20\}$, kept below $\epsilon_{\text{floor}} = 0.2$ (Table 2) to reflect that tolerance cannot drop below the threshold required to speak, and the unfollow/block propensity $P_u \in \{0.05, 0.08, 0.11, 0.15\}$. Each of the $16 \times 2 = 32$ conditions is simulated with $N = 1000$ agents over 40,000 steps — capturing active discourse rather than converged equilibrium, following calibration data [5] — and replicated across $n = 15$ independent network seeds (480 runs total). We compare non-hub posting activity across the three camps (Section 3.1). Code is publicly available on GitHub.[3]

[3] https://github.com/tooooomoya/neutral-hubs-beyond-direct-transmission

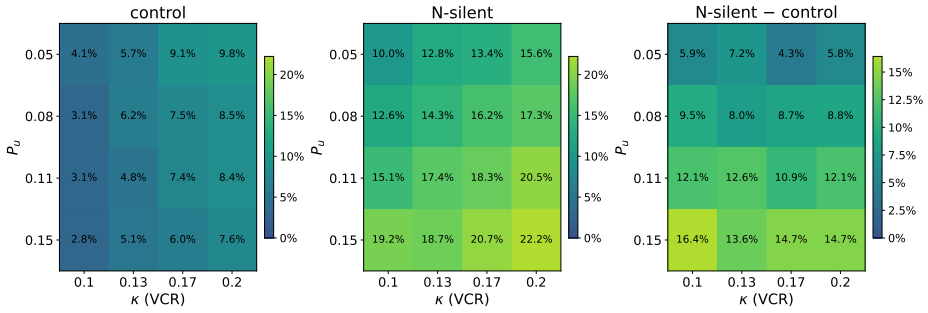


Fig. 2: Late trailing-window share of an extreme agent's feed occupied by same-signed moderate-authored content: for *control* and *N-silent* (left and center), and their difference (right).

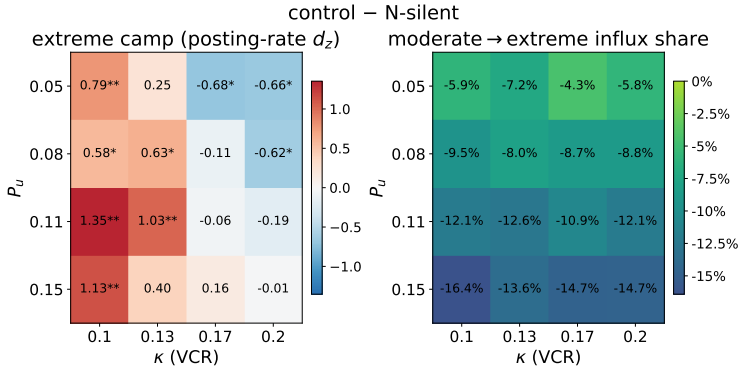


Fig. 3: Same two complimentary effects as Fig. 1 right and Fig. 2 right, with the contrast reversed to *control* − *N-silent*, framing them via the presence of an active neutral hub: extreme-camp posting-rate effect size (left) and moderate→extreme influx share gap (right).

5 Experimental Results

Fig. 1 displays heatmaps of the paired Cohen's d_z (mean seed-wise difference divided by its standard deviation) in per-capita posting rate across the three camps over the trailing window $(38000, 40000]$, comparing *N-silent* against *control* (*N-silent–control*). Neutrals consistently increase posting activity when perceiving aligned hub vocality, whereas moderates reduce theirs. In contrast, the extreme camp exhibits a non-linear response: its posting is reduced (negative d_z) at lower κ values for each P_u row, with the sign-crossing point shifting across P_u levels. Reversing the contrast to *control–N-silent* frames these differences via the presence of neutral hubs (Fig.3 left). From this perspective, for a given P_u, neutral hubs facilitate extreme posting under low comfort thresholds (low κ), and suppress it when higher nearby agreement is required (high κ).

Also, P_u acts as a modulator that shifts the κ sign-crossing point, the watershed at which neutral-hub vocality switches between facilitating and suppressing the extreme camp. In Fig. 1, higher P_u rows maintain negative effect sizes over a broader low-κ range before turning positive, confirming that P_u governs where this sign reversal occurs.

To uncover the upstream mechanism behind the neutral hubs' cross-over effect on extremes, we examine the interplay between moderates and extremes. Fig. 2 displays the trailing-window share of an extreme agent's feed occupied by same-signed moderate content. This share is consistently larger in *N-silent* than in *control*. As shown in the rightmost panel, this asymmetric influx difference is mainly governed by P_u, growing several-fold as P_u increases, and only weakly by κ.

6 Discussion

The uniform responses of neutrals (amplified) and moderates (suppressed) in Fig. 1 follow from a single mechanism: by construction (Eq. (4)), an agent's posting probability depends solely on the comfortable fraction $c_i(t)$ of its feed. Neutral hubs directly move the posting behavior of whichever camp receives their output in substantial volume. For the neutral camp, hub opinions align with the neutral range, placing almost every hub post within the agents' vocal comfort radius κ. Because neutral hubs supply the vast majority of neutral-authored content, silencing them reduces a neutral agent's average feed-comfort fraction from roughly 50% to 30%, driving down its posting activity — indeed, neutral-hub content accounts for nearly 80% of a neutral agent's feed in control conditions, thus the disappearance of this large body of homophilic content is what drives the drop. Conversely, for moderates, neutral hubs occupy 25%–50% of their feeds in control conditions (depending on P_u), but their opinions fall almost entirely outside the moderates' comfort range. Activating neutral hubs thus dilutes the moderates' comfort ratio and suppresses their posting, whereas silencing them removes this dilution and reactivates them.

The extreme camp's response, however, cannot be explained by these direct mechanisms, as neutral-hub content occupies only a few percent of an extreme agent's feed in all control conditions, suggesting that the extreme camp's sign reversal is a downstream ripple effect. We argue this operates through two sequential layers: an *openness* layer (homophily-driven gating) that determines how much moderate content reaches extreme agents, followed by a *comfort* layer (a spiral-of-silence mechanism) that determines whether that arriving content is evaluated as supportive or disruptive.

The openness mechanism is governed by P_u and operates via $\epsilon_i(t)$. Because ties are formed or severed based on $\epsilon_i(t)$, an extreme agent's threshold determines whether a moderate author is followed or blocked. Crucially, this gating is non-uniform: a narrow $\epsilon_i(t)$ admits only the extreme-ward edge of moderates, whereas a wider $\epsilon_i(t)$ extends admission toward their neutral-ward edge. Higher P_u correlates with a wider converged $\epsilon_i(t)$ among extreme agents, a monotonic

trend observed across all κ strata (Kendall's $\tau = 0.29$–0.38, $p < 10^{-4}$). This arises because while $\epsilon_i(t)$ drops initially due to exposure to out-of-tolerance content, a higher P_u prunes distant sources faster, allowing $\epsilon_i(t)$ to recover to a wider threshold.

Supporting this role of P_u, the condition gap in moderate post volume reaching extreme feeds (Fig. 2) grows consistently with P_u. As a network-level validation, the *N-silent–control* gap in mean directed edges from extreme to same-sign moderate agents at $t = 40{,}000$ also increases with P_u (from $+0.08$ at $P_u = 0.05$ to $+0.22$ at $P_u = 0.15$). This directly confirms at the graph level the feed-exposure pattern shown in Fig. 2.

The comfort mechanism, in contrast, is governed by κ. Whatever moderate content the openness mechanism admits, κ dictates its evaluation: content within κ is processed as comfortable, while content outside it acts as disruptive. This evaluation role fixes the direction of the extreme camp's sign reversal. Under a narrow κ, little of the admitted moderate content falls within the comfort range, diluting the comfort ratio and suppressing posting; under a wide κ, a non-trivial share falls inside, boosting posting instead. Thus, for a given P_u, silencing neutral hubs suppresses extreme posting when κ is low, but activates it when κ is high. As shown in Fig. 1, this sign-crossing point shifts toward higher κ as P_u increases because the wider $\epsilon_i(t)$ retains neutral-ward moderate sources, requiring a wider κ for their content to register as comfortable. Notably, this effect weakens at high P_u and high κ, as the wider $\epsilon_i(t)$ admits both comfortable and uncomfortable moderate content, partially canceling each other out.

To capture the specific effect of an *active* neutral hub, we restate this mechanism from the complementary perspective of activation rather than silencing, consequently we show the complimentary heatmap results in Fig. 3. The postive region of Fig. 3 corresponds to the facilitating regime, where an active neutral hub boosts extreme posting, whereas the positive region represents the suppressing regime. In the facilitating regime (low κ), active neutral hubs prevent the arrival of disruptive moderate content that would otherwise enter feeds under *N-silent*, keeping the extreme camp's feed comfortable. In the suppressing regime (high κ), they instead block moderate content that would have satisfied the wider κ, depriving the extreme camp of a comfortable content source. Because P_u adjusts this κ flipping point, an active neutral hub operates less as a passive conduit than as an upstream regulator of posting activity.

7 Conclusion

In this study, we investigated how neutral-hub activity shapes vocalization dynamics in an agent-based model on a heavy-tailed follow network. Neutral hubs exert direct, opposing effects on neutral and moderate populations: they amplify posting among aligned neutrals while suppressing it among moderates. In contrast, their effect on extreme agents is indirect and non-linear, driven by changes in moderate content flows. This downstream effect operates through two layered mechanisms: an *openness* layer that gates moderate content expo-

sure, and a *comfort* layer that evaluates arriving content, causing the net effect on extreme posting to reverse across parameter regimes. Rather than passive bridges connecting distant camps, neutral hubs can act as upstream regulators whose activity reshapes the broader information environment.

These findings should be interpreted within the assumptions of the model. In particular, the observed sign reversal depends on the interaction between adaptive exposure and comfort-driven posting, and the model abstracts opinion into a single dimension while omitting other features of social media, such as recommendation mechanisms and complex interaction with news media. The present results therefore do not imply that neutral-hub activity will necessarily produce the same effects. Nevertheless, within this simplified setting, the observed patterns provide a meaningful indication that neutral-hub activity can generate indirect and counterintuitive downstream effects by reshaping vocality and information flows across the network. Examining whether similar mechanisms emerge under alternative network, behavioral, and platform conditions, and validating these dynamics against empirical data, remains an important direction for future work.

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